

Feature Space Specification

163-Dimensional State Representation for the GBT Directional Classifier
and PPO Reinforcement Learning Agent

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1 Introduction

The directional GBT classifier and the PPO reinforcement learning agent both operate on a shared 163-dimensional feature vector $\mathbf{x}_t \in \mathbb{R}^{163}$, computed at each 5-minute bar t during regular trading hours (09:30–16:00 ET). This vector encodes the complete market microstructure state as seen from the perspective of a 0DTE options market participant: spot price dynamics, the full Greek surface, dealer inventory exposures, volatility term structure, cross-asset context, and temporal position within the trading session.

The feature set is defined in `hybrid_model.py` under the constant `FEATURE_COLUMNS`, which is imported identically by the GBT training pipeline, the RL episode generator, and the live inference bot. This guarantees perfect train/test/production parity.

Definition 1 (Feature Vector). *The feature vector at time t is the ordered tuple:*

$$\mathbf{x}_t = (x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(163)}) \in \mathbb{R}^{163} \quad (1)$$

where each component $x_t^{(i)}$ is a real-valued statistic computed from publicly observable market data up to and including time t . No lookahead is permitted: $x_t^{(i)} \perp \{S_\tau : \tau > t\}$.

2 Feature Taxonomy

The 163 features are organized into ten functional categories, summarized in Table 1.

Category	Primary Information Content	Dimensions
Spot & Log Returns	Price level, 1/5/15/30-min returns, vol-adj returns	8
Technical Indicators	RSI, MACD, Bollinger Bands, momentum	10
Greek Exposures	$\Delta, \Gamma, \theta, \mathcal{V}, \rho$ by expiry bucket	24
GEX / DEX / VEX	Aggregate gamma, delta, vanna exposure measures	18
Put/Call Metrics	Volume PCR, OI PCR, delta-filtered PCR	12
IV Surface	ATM IV, skew, term structure, call/put smile slopes	16
Order Flow Derivatives	Net gamma, net vanna, net charm, DGEX, cross-products	14
Level Proximity	Distance to max-gamma strike, IB levels ($D-1$ to $D-5$)	21
Temporal Encoding	Time-of-day sinusoid, day-of-week sinusoid, minutes-to-close	5
Cross-Ticker Context	QQQ/VIX/TLT returns, spread, correlation	35
Total		163

Table 1: Summary of feature categories and their dimensionality.

3 Category Definitions

3.1 Spot Price and Log Returns ($D = 8$)

The raw spot price S_t is standardized relative to its recent distribution:

$$x_t^{\text{spot}} = \frac{S_t - \bar{S}_{20}}{s_{20}}, \quad \bar{S}_{20} = \frac{1}{20} \sum_{i=1}^{20} S_{t-i}, \quad s_{20} = \text{std}\{S_{t-i}\}_{i=1}^{20} \quad (2)$$

Multi-horizon log returns are included at four lookback windows $\Delta \in \{1, 5, 15, 30\}$ minutes:

$$r_{t,\Delta} = \log\left(\frac{S_t}{S_{t-\Delta}}\right) \quad (3)$$

Volatility-adjusted counterparts normalize each return by the implied 0DTE ATM volatility σ_t^{ATM} scaled to the corresponding horizon:

$$\tilde{r}_{t,\Delta} = \frac{r_{t,\Delta}}{\sigma_t^{\text{ATM}} \sqrt{\Delta/(252 \times 390)}} \quad (4)$$

This transformation maps raw price dynamics into a stationary, approximately standard-normal space, conditioning the GBT on moves expressed in implied-volatility units rather than raw basis points.

3.2 Technical Indicators ($D = 10$)

Classical price-based indicators are included as non-linear compressors of recent price history:

- **RSI(14):** 14-period Relative Strength Index, bounded in $[0, 100]$.
- **MACD and signal:** 12-period minus 26-period exponential moving averages, plus the 9-period signal line and histogram.
- **Bollinger Bands:** upper band $\mu_{20} + 2\sigma_{20}$, lower band $\mu_{20} - 2\sigma_{20}$, and bandwidth $4\sigma_{20}/\mu_{20}$.

3.3 Greek Exposures ($D = 24$)

Aggregate net Greeks across the listed options chain are computed for both 0DTE and the nearest weekly expiration:

$$\mathcal{G}_t^{(k)} = \sum_{\substack{i \in \text{chain} \\ T_i = T_k}} \text{OI}_i \cdot \text{Greek}_i, \quad k \in \{0\text{DTE}, \text{weekly}\} \quad (5)$$

where OI_i is the open interest of contract i . Six Greeks ($\Delta, \Gamma, \theta, \mathcal{V}, \rho, \text{charm}$) are computed for each of the two expiry buckets, yielding 12 features per side (24 total).

3.4 GEX, DEX, and VEX ($D = 18$)

Dealer exposure measures estimate the aggregate hedging pressure that market makers exert on the spot price:

Definition 2 (Gamma Exposure (GEX)). *The market-maker net gamma exposure at time t is:*

$$GEX_t = \sum_{i \in \text{chain}} \Gamma_i \cdot OI_i \cdot \text{Multiplier} \cdot S_t^2 / 100 \quad (6)$$

A positive GEX_t indicates dealers are net long gamma: they sell rallies and buy dips, imposing mean-reverting pressure on the spot price. Negative GEX indicates dealers are net short gamma, amplifying directional moves.

Definition 3 (Delta Exposure (DEX) and Vanna Exposure (VEX)).

$$DEX_t = \sum_i \Delta_i \cdot OI_i \cdot \text{Multiplier} \cdot S_t \quad (7)$$

$$VEX_t = \sum_i \mathcal{V}_i^{(\text{vanna})} \cdot OI_i \cdot \text{Multiplier} \quad (8)$$

DEX measures the directional delta hedge flow. VEX captures the sensitivity of dealer delta to changes in implied volatility, becoming particularly relevant during VIX spike regimes.

These three exposure metrics are computed separately for 0DTE, weekly, and the combined chain, yielding 18 features.

3.5 Put/Call Metrics ($D = 12$)

Put/call ratios provide an aggregate measure of options market sentiment:

$$\text{PCR}_{\text{vol}} = \frac{V_{\text{put}}}{V_{\text{call}}} \quad (9)$$

$$\text{PCR}_{\text{OI}} = \frac{OI_{\text{put}}}{OI_{\text{call}}} \quad (10)$$

$$\text{PCR}_{\Delta} = \frac{\sum_{i: \Delta_i < 0} |\Delta_i| \cdot OI_i}{\sum_{j: \Delta_j > 0} \Delta_j \cdot OI_j} \quad (11)$$

where V denotes traded volume. The delta-filtered PCR weights contracts by their hedge ratio, providing a more precise measure of directional pressure than raw volume ratios. These are computed for three expiry tiers (0DTE, weekly, all).

3.6 Implied Volatility Surface ($D = 16$)

The IV surface features encode the term structure and cross-sectional shape of implied volatility:

- **ATM IV (0DTE, weekly, monthly):** implied volatility at the at-the-money strike.
- **IV Skew:** $IV_{25\Delta, \text{put}} - IV_{25\Delta, \text{call}}$, a risk-reversal proxy for demand asymmetry.
- **Term structure:** weekly/0DTE IV ratio and monthly/weekly IV ratio.
- **Smile slopes:** OTM call slope and OTM put slope, capturing tail-risk premium asymmetry.

3.7 Order Flow Derivatives ($D = 14$)

Higher-order Greek cross-products capture the sensitivity of dealer hedge flows to joint price-volatility moves:

$$\text{net_vanna}_t = \sum_i \frac{\partial^2 C_i}{\partial S \partial \sigma} \cdot \text{OI}_i \quad (12)$$

$$\text{net_charm}_t = \sum_i \frac{\partial^2 C_i}{\partial S \partial t} \cdot \text{OI}_i \quad (13)$$

$$\text{DGEX}_t = \left. \frac{d(\text{GEX})}{dS} \right|_t \quad (14)$$

These terms become significant near zero-day expiration when the term-structure derivatives of gamma approach infinity.

3.8 Level Proximity and Initial Balance ($D = 21$)

Market microstructure research has long identified that traders react asymmetrically to the initial price range established during the first hour of trading. The Initial Balance (IB) for day d is:

$$\text{IB}_{\text{high}}^{(d)} = \sup_{\tau \in [09:30, 10:30]_d} S_\tau, \quad \text{IB}_{\text{low}}^{(d)} = \inf_{\tau \in [09:30, 10:30]_d} S_\tau \quad (15)$$

Distance features quantify the spot price's position relative to IB levels for the current and prior five trading days $d \in \{0, -1, -2, -3, -4, -5\}$:

$$\delta_{\text{high}}^{(d)} = \text{clip} \left(\frac{S_t - \text{IB}_{\text{high}}^{(d)}}{\nu \cdot S_t}, -10, 10 \right), \quad \nu = 0.005 \quad (16)$$

An inside-IB boolean $\mathbf{1}[\text{IB}_{\text{low}}^{(0)} \leq S_t \leq \text{IB}_{\text{high}}^{(0)}]$ is also included.

Additionally, confluence features encode whether the current spot price aligns with the maximum-gamma strike K_Γ^* via a Gaussian RBF kernel:

$$\Phi_t^\Gamma = \exp \left(-\frac{(S_t - K_\Gamma^*)^2}{2(0.005 \cdot S_t)^2} \right), \quad K_\Gamma^* = \arg \max_K \sum_{i: K_i = K} \Gamma_i \cdot \text{OI}_i \quad (17)$$

$\Phi_t^\Gamma \rightarrow 1$ when spot approaches the dominant gamma strike; $\Phi_t^\Gamma \rightarrow 0$ in open price space. Analogous confluence features are computed for Vanna and DGEX strikes.

3.9 Temporal Encoding ($D = 5$)

Intraday and weekly periodicity is encoded via Fourier projections to prevent artificial discontinuities at the day and week boundaries that would otherwise confuse tree-based splits:

$$x_{\text{tod}}^{\sin} = \sin \left(2\pi \cdot \frac{m_t}{390} \right), \quad x_{\text{tod}}^{\cos} = \cos \left(2\pi \cdot \frac{m_t}{390} \right) \quad (18)$$

$$x_{\text{dow}}^{\sin} = \sin \left(2\pi \cdot \frac{d_t}{5} \right), \quad x_{\text{dow}}^{\cos} = \cos \left(2\pi \cdot \frac{d_t}{5} \right) \quad (19)$$

where $m_t \in [0, 390]$ is the number of minutes elapsed since the 09:30 open, and $d_t \in \{0, 1, 2, 3, 4\}$ is the day-of-week index. A raw `minutes_to_close` = $390 - m_t$ feature is also included to allow the GBT to model the accelerating theta-decay regime in the final hour.

3.10 Cross-Ticker Context ($D = 35$)

The system tracks four secondary instruments: QQQ, VIX, TLT, and the SPX/XXX spread. For each, multi-horizon returns and selected Greeks are included:

- **VIX:** spot level, 1-min and 5-min returns. Acts as an implied-volatility regime indicator.
- **TLT (20Y Treasury ETF):** 1-min, 5-min, and 15-min returns. Captures risk-off flows.
- **XXX:** full analog of the SPX feature vector (spot, returns, Greeks). Encodes tech-sector divergence.
- **SPX/XXX spread:** normalized difference in 5-minute returns. Captures sector rotation.

4 Numerical Preprocessing Pipeline

4.1 Log Transform for Extreme-Scale Features

Features with raw magnitudes exceeding 10^4 (e.g., raw GEX in dollar-gamma units) undergo the sign-preserving log transform before being written to the training Parquet:

$$\tilde{x} = \text{sign}(x) \cdot \log(1 + |x|) \quad (20)$$

This is applied to: `net_gamma`, `net_vanna`, `net_charm`, `net_dgex`, `gex_0dte`, `gex_weekly`, `delta_gamma_ratio`, and all raw exposure scalars.

4.2 Feature Normalizer

The `FeatureNormalizer` class fits per-feature mean μ_j and standard deviation σ_j on the training window:

$$x_{\text{norm}}^{(j)} = \frac{x^{(j)} - \mu_j}{\sigma_j + \varepsilon}, \quad \varepsilon = 10^{-8} \quad (21)$$

Statistics are fit once per walk-forward window and serialized to `models/hybrid_normalizer_wf.npz`. At inference time, the normalizer is loaded and applied identically to every input batch, including the live trading bot.

4.3 NaN Handling

The GBT training pipeline applies `np.nan_to_num(X, nan=0.0, posinf=5.0, neginf=-5.0)` before model fitting. LightGBM natively handles missing values (NaN) via its optimal split-finding mechanism, which learns a dedicated branch direction for NaN samples. The explicit imputation to zero is applied only as a safety measure for features that should never be NaN in well-formed data.

5 Feature Importance and Selection

LightGBM provides three native feature importance metrics: split count, gain, and permutation importance. Across walk-forward windows, the consistently highest-importance features are:

Rank	Feature	Category	Gain Contribution
1	vix_spot	Cross-Ticker	High
2	ret_5m_vol_adj	Spot & Returns	High
3	gex_0dte	GEX/DEX/VEX	Medium-High
4	pcr_vol_0dte	Put/Call	Medium-High
5	iv_skew_0dte	IV Surface	Medium-High
6	dist_ib_high	Level Proximity	Medium
7	net_vanna	Order Flow	Medium
8	minutes_to_close	Temporal	Medium
9	confluence_max_gamma	Level Proximity	Medium
10	tlr_ret_5m	Cross-Ticker	Medium

Note: Rankings are window-averaged; individual windows may vary.

Table 2: Top-10 features by average LightGBM gain across walk-forward windows.

Remark 1. *The dominance of `vix_spot` and `ret_5m_vol_adj` confirms that volatility regime and recent momentum are the primary drivers of 0DTE directional predictability. The GEX and IB features provide secondary conditioning, particularly for mean-reversion signals near dominant gamma strikes.*

6 RL State Space Extension

The PPO agent observes an extended 182-dimensional state vector $\mathbf{s}_t$ that appends 19 dynamic and position-management features to the base 163-dimensional market feature vector:

$$\mathbf{s}_t = \left[\mathbf{x}_t \mid \mathbf{x}_t^{\text{dyn}} \mid \mathbf{q}_t \mid \mathbf{c}_t \right] \in \mathbb{R}^{182} \quad (22)$$

where:

- $\mathbf{x}_t^{\text{dyn}} \in \mathbb{R}^8$ Dynamic market features: per-minute spot history, delta history, and volume.
- $\mathbf{q}_t \in \mathbb{R}^7$ Position state: unrealized PnL%, hold time, $|\Delta|$, recovery probability, implied volatility, maximum adverse excursion (MAE), and trailing drawdown.
- $\mathbf{c}_t \in \mathbb{R}^4$ Model context: GBT confidence score c_t , time-to-target, dummy edge indicator, and $\log \sigma$ (observation noise).

The state is augmented during training via `augment_state()`, which adds small Gaussian noise $\mathcal{N}(0, 0.01)$ to all continuous components to improve policy generalization.

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